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SCHC Compression of VOICI Headers

Abstract

This document describes how to compress VOICI headers using SCHC rules at a lower stratum, demonstrating that VOICI provides explicit routing metadata that is transparently compressed to minimal on-wire size while preserving standardized dispatch semantics. It is intended as a reference for implementers deploying VOICI in SCHC-based networks and does not modify the VOICI specification itself.

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1. Introduction

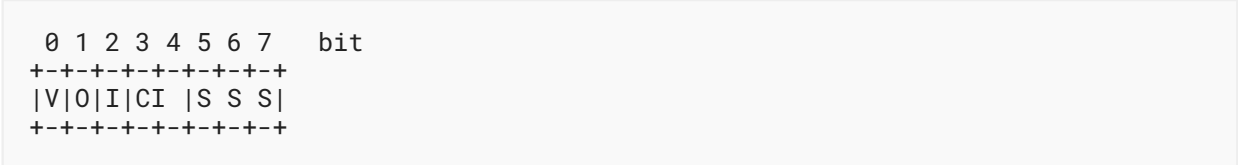
The VOICI link multiplexer ~[VOICI] provides a standardized transport header that carries session identification, content mechanism dispatch, and optional integrity protection with a minimal 1-byte header in the common case.

When VOICI is carried over a link where SCHC is also used for header compression, a lower SCHC stratum can compress the VOICI header itself. This document describes the SCHC rules that achieve this compression and demonstrates how VOICI provides explicit routing metadata that is transparently compressed, achieving minimal on-wire overhead while preserving standardized dispatch semantics.

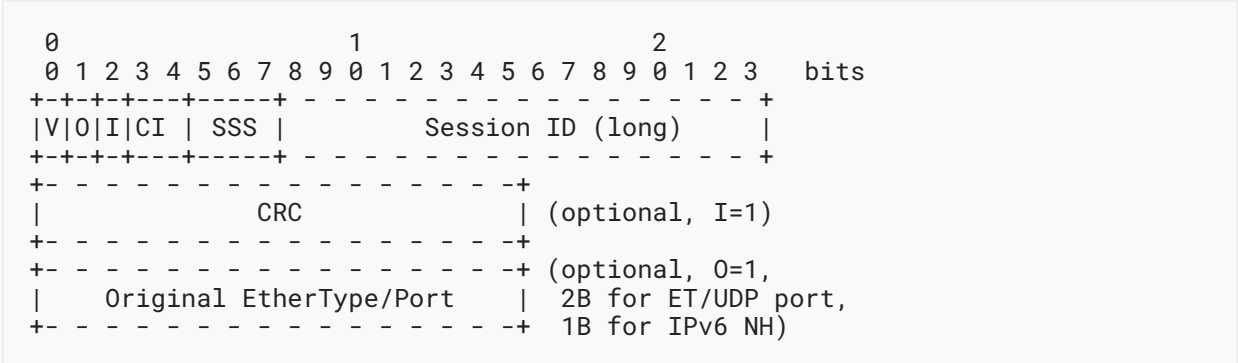
The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

2. VOICI Header Layout

The VOICI header is defined in ~[VOICI]. For reference, the minimal header (no optional fields, inline Session ID) is a single byte:



The full header, including all optional fields, is:



The first byte is always present. The V-O-I flags (3 bits), CI field (2 bits), and SSS field (3 bits) form the mandatory header. The optional fields are present only when their corresponding flags are set:

- **V** (1 bit): Header version. V=0 for the current specification.

- **O** (1 bit): Original EtherType/Port flag. When set, the Original EtherType/Port field is present at the end of the header.
- **I** (1 bit): Integrity flag. When set, a 2-byte CRC-16 field is present after the Session ID.
- **CI** (2 bits): Content Identifier. CI=0 for unprocessed/raw, CI=1 for SCHC, CI=2 reserved, CI=3 for extended CI.
- **SSS** (3 bits): Session ID prefix. Values 0-6 represent inline Session IDs; value 7 triggers a LEB128 extension in the following byte(s) with offset +7.
- **Session ID (long)**: Variable-length LEB128-encoded Session ID, present only when SSS=7 (or when CI=3 for extended CI).
- **CRC**: 16-bit CRC-16 integrity check, present only when I=1.
- **Original EtherType/Port**: 1-2 bytes carrying the original EtherType, IPv6 Next Header, or UDP port number, present only when O=1.

For the common case (V=0, O=0, I=0, CI=1 for SCHC, SSS=0 for Session 0), the byte value is:

```
V=0, O=0, I=0, CI=01, SSS=000  =>  0b00001000  =>  0x08
```

For Session 1 (SSS=1), the byte value is 0x09. The full range of inline sessions (0-6) with CI=0b001 (SCHC Content) maps to bytes 0x08 through 0x0E.

3. SCHC Compression Rules for VOICI

3.1. Field Identifiers

To write SCHC compression rules for the VOICI header, the following Field Identifiers (FIDs) are defined. These FIDs reference specific sub-fields of the VOICI header byte, as well as optional extension fields:

FID	Field	Size
VOICI.V	Version	1 bit
VOICI.O	Original ET/Port flag	1 bit
VOICI.I	Integrity flag	1 bit
VOICI.CI	Content Identifier	2 bits
VOICI.SSS	Session ID prefix	3 bits
VOICI.SID	Session ID (LEB128)	variable
VOICI.EXT_CI	Extended CI (LEB128)	variable
VOICI.CRC	CRC-16	16 bits

FID	Field	Size
VOICI.ORIG_EP	Original ET/Port	1-2 bytes

Table 1

The per-field FIDs (VOICI.V through VOICI.SSS) enable SCHC compression rules that elide known sub-fields while preserving the Session ID identifier required for dispatch, original EtherType/ NextHeader/Port and CRC when present.

3.2. Scenario A: Single Session (0-bit residue) with minimal VOICI header

These examples illustrate a stratum scoped to the VOICI header alone. If the stratum spans multiple headers (e.g., `[IPv6 | VOICI]`), additional Field Descriptors are required for those headers (as shown in the concrete example). When only one session exists on the link, the VOICI header has fully deterministic content, and per-field rules elide each sub-field individually.

```
+-----------+--+---+--+-----+-------+----------+
|     FID   |FL| FP|DI| TV  |  MO   |    CDA   |
+-----------+--+---+--+-----+-------+----------+
| VOICI.V   | 1| 1 |Bi| 0   | equal | not-sent |
| VOICI.O   | 1| 1 |Bi| 0   | equal | not-sent |
| VOICI.I   | 1| 1 |Bi| 0   | equal | not-sent |
| VOICI.CI  | 2| 1 |Bi| 0x01| equal | not-sent |
| VOICI.SSS | 3| 1 |Bi| 0   | equal | not-sent |
+-----------+--+---+--+-----+-------+----------+

Residue: 0 bits (all fields elided)
```

3.3. Scenario B: Two Sessions (1-bit residue)

When two sessions are multiplexed -- Session IDs 0 and 1 mapped to SSS values 0b000 and 0b001, the session identifier must be preserved. A 1-bit residue is achieved by eliding the known flag and CI bits while preserving SSS via MSB matching:

```
+-----------+--+---+--+-----+-------+----------+
|     FID   |FL| FP|DI| TV  |  MO   |    CDA   |
+-----------+--+---+--+-----+-------+----------+
| VOICI.V   | 1| 1 |Bi| 0   | equal | not-sent |
| VOICI.O   | 1| 1 |Bi| 0   | equal | not-sent |
| VOICI.I   | 1| 1 |Bi| 0   | equal | not-sent |
| VOICI.CI  | 2| 1 |Bi| 0x01| equal | not-sent |
| VOICI.SSS | 3| 1 |Bi| 0   | MSB(2)|    LSB   |
+-----------+--+---+--+-----+-------+----------+

Session 0 (SSS=0b000) -> MSB(2) match -> residue: 0
Session 1 (SSS=0b001) -> MSB(2) match -> residue: 1

Residue: 1 bit
```

3.4. Extended Scenarios

3.4.1. Multiple Sessions with Inline IDs

For up to 7 inline sessions (SSS values 0-6), the per-field approach with MSB matching on SSS can encode any number of sessions:

- 2 sessions (SSS 0-1): MSB(2) match -> 1 bit residue
- 3-4 sessions (SSS 0-3): MSB(1) match -> 2 bits residue
- 5-7 sessions (SSS 0-6): Ignore/Value-Sent -> 3 bits residue

3.4.2. Sessions Beyond Inline Range

For sessions with SSS=7 (LEB128 extension), the compressor must handle the variable-length Session ID field. A VOICI-aware compressor can preserve the LEB128-encoded Session ID as a residual field, producing a residue proportional to the Session ID magnitude (1 additional byte for Session IDs 7-134, 2 additional bytes for 135-65535).

4. Concrete Example: Multi-Stratum Deployment

This section illustrates a deployment using multiple SCHC strata and VOICI for inter-stratum session multiplexing. It demonstrates how VOICI provides explicit routing metadata that is transparently compressed by a lower stratum, achieving minimal on-wire overhead while preserving standardized dispatch semantics.

4.1. Deployment Scenario

The deployment consists of a 6Lo mesh network with a 6LN (end device), a 6LR (mesh router), and a 6LBR (border router). Beyond the 6LBR, a Server hosting both CoAP and CoAPS resides on the Internet.

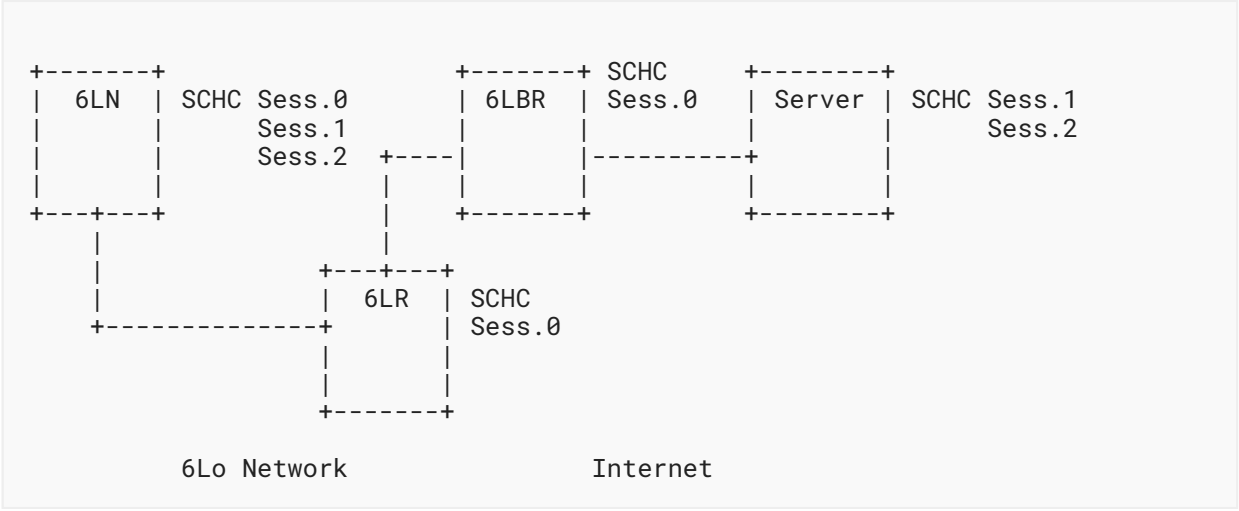
The 6LN sends unencrypted CoAP traffic to the Server on UDP port 5683 and encrypted CoAPS traffic to the same Server on UDP port 5684.

4.1.1. SCHC Considerations

The 6LN participates in 3 SCHC Sessions, Sess.0, Sess.1 and Sess.2.

The 6LR and 6LBR participate in the SCHC Session Sess.0 only and the Server participates in Sessions Sess.1 and Sess.2.

- Sess.1 and Sess.2 and corresponding Contexts are managed by a first actor, for instance a utility company that operates the Server. Both are executed on the same Endpoint that uses VOICI as the Dispatcher.
- Sess.0 and corresponding Context is managed by the Network Operator of the 6Lo Network. Having a single Session, the corresponding Endpoint dispatches SCHC Datagrams directly to the appropriate Instance.



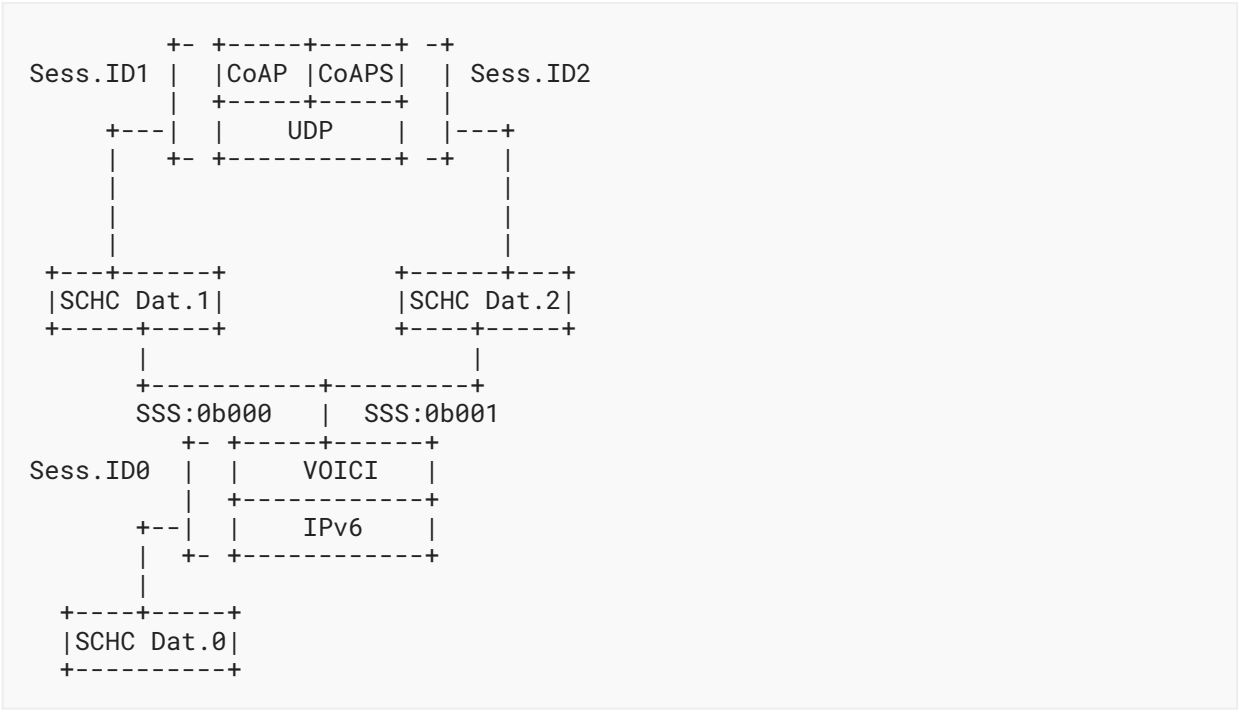
The deployment uses three SCHC strata:

- **Upper Stratum 1**: Compresses `[UDP | CoAP]` in Session 1
- **Upper Stratum 2**: Compresses `[UDP | CoAPS]` in Session 2
- **Lower Stratum**: Compresses `[IPv6 | VOICI]` in Session 0

4.2. SCHC processing at the 6LN

The following figure illustrates the 3 SCHC Strata and 3 Instances on the 6LN host.

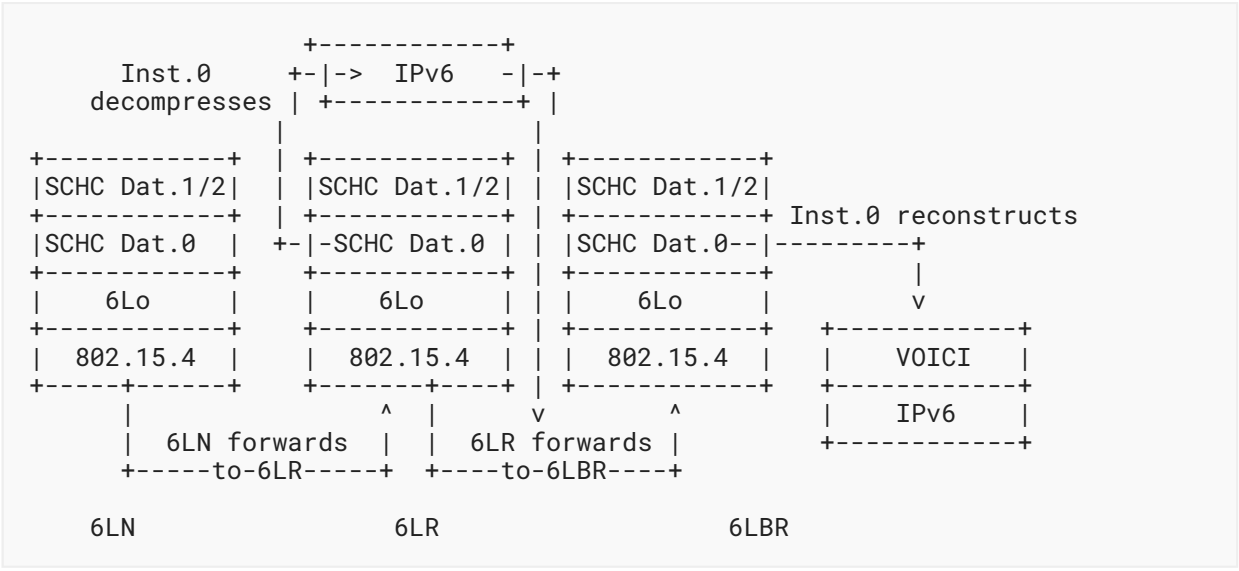
1. Inst.1 compresses UDP/CoAP and produces the SCHC Datagram SCHC Dat.1 Inst.2 compresses UDP/CoAPS and produces the SCHC Datagram SCHC Dat.2
2. VOICI encapsulates SCHC Dat.1 with SSS:=0b000 or SCHC Dat.2 with SSS:=0b001. IPv6 Next Header is set to SCHC.
3. Inst.0 compresses IPv6/VOICI and produces SCHC Dat.0 VOICI.SSS MO/CDA set to msb/lsb and TV set to 0b00



4.3. SCHC Datagrams on the 6Lo Network

The following Figure illustrates how the datagrams are transported over the 6Lo network.

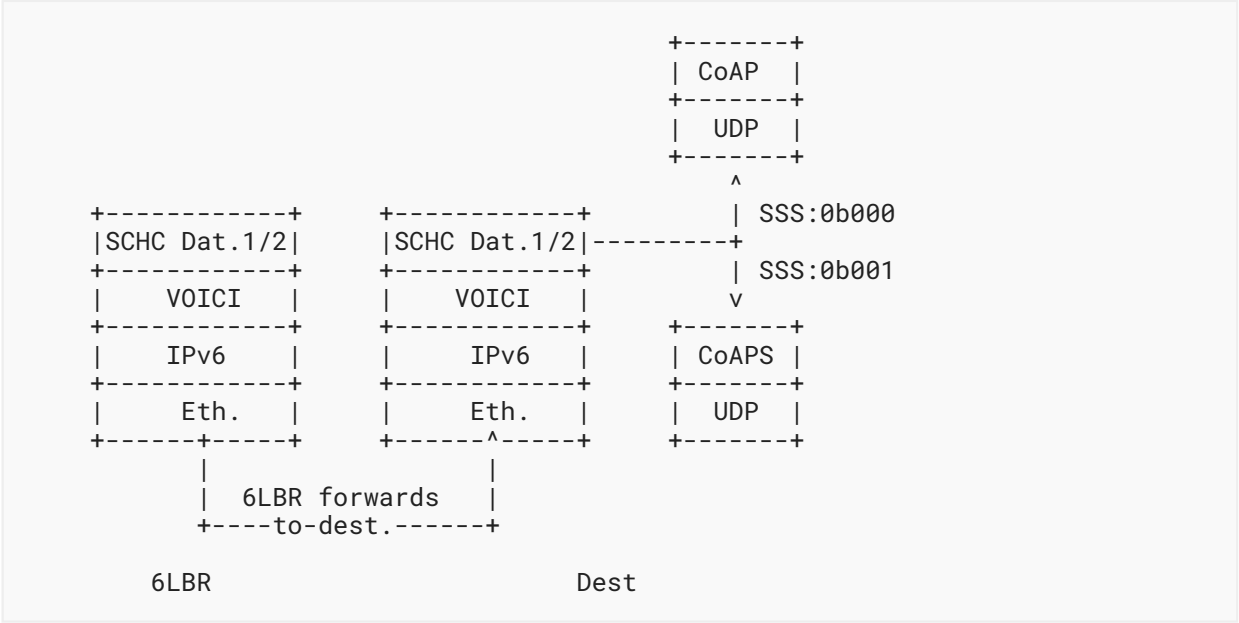
1. 6LN encapsulates the resulting datagram, that is the concatenation of SCHC Dat.0 and SCHC Dat.1/2, in the 802.15.4 payload with the SCHC Dispatch byte set and sends it to 6LR.
2. 6LR detects the SCHC dispatch byte and dispatches the frame to Inst.0. Inst.0 decompresses SCHC Dat.0, retrieves IPv6 header (and VOICI) and forwards to 6LBR.
3. 6LBR detects the SCHC dispatch byte and dispatches the frame to Inst.0. Inst.0 decompresses SCHC Dat.0, reconstructs IPv6 and VOICI headers. 6LBR forwards the IPv6 frame to its destination, outside of the 6Lo network.



4.4. Transport over the Internet and processing at the Server

The following Figure illustrates the SCHC datagrams transported over the Internet in IPv6 frames and how they are processed at the Server.

1. 6LBR forwards to Server
2. The Server Endpoint dispatches the SCHC Dat. to Inst.1/2 based on VOICI.SSS value. SCHC Datagrams identified by VOICI.SSS = 0b000 are dispatched to Inst.1 which decompresses UDP and CoAP and those with VOICI.SSS= 0b001 are dispatched to Inst.2 which decompresses UDP and CoAPS.
3. The Original IPv6 Next Header value is restored to its original value (UDP,17) and the frame is handled by UDP.
4. The CoAP(S) server processes the message.



4.5. Formats

Before any compression, the packet contains standard IPv6/UDP/CoAP with the following IPv6 addresses, fd00:1::4 for the 6LN and 2001:db8::13 for the Server.

4.5.1. Upper Stratum Compression

Each upper stratum compresses its respective UDP/CoAP headers independently. After upper compression, VOICI encapsulates the upper SCHC datagram with a Session ID that routes it to the correct upper context at the receiver:

- **Session 1**: VOICI byte = `0x08` (V=0, O=0, I=0, CI=01, **SSS=000**)
- **Session 2**: VOICI byte = `0x09` (V=0, O=0, I=0, CI=01, **SSS=001**)

The IPv6 Next Header field is set to indicate the presence of SCHC-compressed data (i.e., the SCHC IP Protocol Number).

The intermediate packet handed to the lower stratum looks like:

```
[CoAP Payload]
[Upper SCHC Residue]
[VOICI Header (SSS=0 or SSS=1)]
[IPv6 Header (NH=SCHC)]
```

4.5.2. Lower Stratum Compression

The lower stratum compresses the IPv6 and VOICI headers jointly using a single Rule. With only one lower stratum Rule, the Rule ID is zero-length; discrimination between the two upper sessions is carried in the 1-bit residue from the VOICI.SSS field (MSB(2) with LSB CDA).

Rule (Session 0)

```
+-----------------+-----+---+--+--------------+--------+----------+
|      FID        |  FL| FP |DI|      TV      |   MO   |    CDA   |
+-----------------+-----+---+--+--------------+--------+----------+
| IPv6 Ver        |   4 | 1 |Bo|      6       | equal  | not-sent |
| IPv6 Traffic Cl.|   8 | 1 |Bo|      0       | equal  | not-sent |
| IPv6 Flow Label |  20 | 1 |Bo|      0       | equal  | not-sent |
| IPv6 Length     |  16 | 1 |Bo|              | ignore | compute-*|
| IPv6 Next Header|   8 | 1 |Bo|   \<SCHC\>   | equal  | not-sent |
| IPv6 Hop Limit  |   8 | 1 |Bo|      64      | equal  | not-sent |
| IPv6 Src Addr   |  64 | 1 |Bo|   fd00:1::4  | equal  | not-sent |
| IPv6 Dst Addr   |  64 | 1 |Bo| 2001:db8::13 | equal  | not-sent |
| VOICI.V         |   1 | 1 |Bi|      0b0     | equal  | not-sent |
| VOICI.O         |   1 | 1 |Bi|      0b0     | equal  | not-sent |
| VOICI.I         |   1 | 1 |Bi|      0b0     | equal  | not-sent |
| VOICI.CI        |   2 | 1 |Bi|     0b01     | equal  | not-sent |
| VOICI.SSS       |   3 | 1 |Bi|     0b00     | MSB(2) |   LSB    |
+-----------------+-----+---+--+--------------+--------+----------+

  Residue: 1 bit
```

4.5.3. On-the-Wire Result

The resulting 802.15.4 payload is: ~~~ +-------------+-----------------+-------------------------------+ |SCHC Dispatch|Lower SCHC Residue|Upper SCHC Residue| CoAP Payload | | (01000100) | (1bit) | | (raw) | +-------------+-----------------+-------------------------------+ ~~~

5. Security Considerations

The SCHC compression of VOICI headers does not introduce new security considerations beyond those of the base VOICI specification ~[VOICI] and SCHC ~[RFC8724]. The compression rule described in this document preserves the integrity of the VOICI Header.

6. IANA Considerations

This document has no IANA actions. The Field Identifiers defined herein are informational references for implementers writing SCHC compression rules for VOICI headers.

7. References

7.1. Normative References

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, DOI 10.17487/RFC2119, March 1997, <https://www.rfc-editor.org/rfc/rfc2119>.

[RFC8174] Leiba, B., "Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words", BCP 14, RFC 8174, DOI 10.17487/RFC8174, May 2017, <https://www.rfc-editor.org/rfc/rfc8174>.

[RFC8724] Minaburo, A., Toutain, L., Gomez, C., Barthel, D., and JC. Zuniga, "SCHC: Generic Framework for Static Context Header Compression and Fragmentation", RFC 8724, DOI 10.17487/RFC8724, April 2020, <https://www.rfc-editor.org/rfc/rfc8724>.

7.2. Informative References

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